

Unequal Skills

10.1 INTRODUCTION

In this chapter, we present and study a new model, which is a variant of the private good production model of the previous chapter. Compared with that model, we assume that the production set is linear and that the agents may have different production skills. Consequently, the agents are characterized by two parameters: their preferences and their skills. The natural interpretation of the model, which was introduced in the literature a long time ago, from different viewpoints, by Mirrlees (1971) and Pazner and Schmeidler (1974), is that input is the labor time of the agents and their skill is the wage rate at which they are able to find a job. This is the canonical setting in which labor income taxation issues can be addressed. In the next chapter, we show how some of the social ordering functions we derive here can be used to design optimal income tax schemes.

Productive skills are assumed to be fixed and independent of the labor time. This amounts to assuming that agents use a linear technology. In view of such linearity, the *laissez-faire* allocations – that is, the allocations at which agents choose their labor time and consume the share of the production that corresponds to their share in the contribution – are natural candidates to qualify as equitable allocations. Indeed, each agent freely chooses his or her preferred bundle and there is no externality among agents.

One can argue, however, that these allocations are inequitable, in particular if differences in skills come (at least partly) from inherited features that cannot be attributed to the agents' responsibility, or if agents could be more productive but are constrained by the unavailability of jobs with higher wage rates.¹ In this chapter, we endorse the viewpoint that, in the presence of unequal skills, *laissez-faire* is unfair, and we study the fairness objectives of (1) neutralizing the consequences of differential skills and (2) being neutral with respect to the preferences – that is, not correcting for different choices of labor time.

¹ It may be, however, that agents' skills are also partly the outcome of previous personal choices about investment in human capital. Symmetrically, one may argue that agents are only partly responsible for their preferences over consumption and leisure.

We believe that these two objectives capture some basic ingredients of the classical debate about labor income taxation. Some people argue that differences in productive skills call for redistribution, whereas others claim that hardworking agents should benefit from their hard work. The social ordering functions we propose in this chapter show possible ways out of the dilemma, and the study of income tax in the next chapter shows how tax schemes can be related to particular ways of finding a compromise between the two major fairness objectives defined in the previous paragraph.

Let us consider these two objectives more closely. To capture the idea that differential skills should not entail unequal individual outcomes, let us consider allocating resources between two agents who have the same preferences. As they differ only with respect to their skills, society should help them reach the same outcome, in terms of well-being. We call it the *objective of compensation*.

To capture the idea that the various preferences displayed in the population should not be treated differently – that is, that the agents should not receive any differential amount of resources on the pure basis that they have “good” or “bad” preferences – let us consider allocating resources between two agents who have the same skill. As they differ only with respect to their preferences, it is natural to consider that society should submit them to the same treatment in terms of redistribution. This implies that, ideally, agents with identical skills should be submitted to the same lump-sum transfer that does not depend on their preferences and their choice of labor. In such a situation, two agents with identical skills have the same linear budget set with a slope equal to their skill. If that is achieved, when an agent changes preferences, one additional labor time unit results in an increased consumption by exactly the corresponding additional amount of production and society is not concerned with the agent’s change in preferences. We call it the *objective of responsibility*.²

The main lesson of this chapter, from the point of view of fairness, is that there exists a deep conflict between these two objectives. One cannot simultaneously say that well-being should not depend on skills when agents have identical preferences, and that transfers should not depend on preferences when agents have identical skills. Given the richness of the model, we are able to propose different axioms of either compensation or responsibility, and check for their mutual compatibility problems. The results of our study are the characterizations of families of social ordering functions that offer reasonable compromises between the two objectives, that is, that satisfy combinations of appropriate weakenings of our basic requirements.

Many fair allocation problems involve agents having different needs, talents, handicaps, and so forth – that is, untransferable attributes that determine how successful they may be in transforming external resources into personal outcomes. A widespread ethical view on those matters is that external resources

² This is called “liberal reward” by Fleurbaey (2008), who states that two different conceptions of the implications of responsibility for reward schemes are distinguished, the “liberal” and the “utilitarian.” We focus here on the former.

must be allocated or reallocated among these agents in a differentiated way to counterbalance such differences in personal attributes.³ We conceived this chapter and the following one to illustrate how to address these questions with the help of social ordering functions.

10.2 THE MODEL

The model of this chapter (and the next one) is a variant of the model we studied in Sections 9.2 through 9.4. Individual bundles are composed of a quantity of labor time (ℓ) and a level of consumption (c). We further add the restriction that labor time is bounded above by a “full time” level ($\ell \leq 1$). The new and key ingredient of the current analysis is the agent’s *production skill* $s_i \geq 0$, enabling the agent to produce the quantity $s_i \ell_i$ of consumption good with labor time ℓ_i . We further assume that $s_i \in S = [s_{\min}, s_{\max}]$, where $s_{\min}$ denotes the lowest possible production skill in society, or, in some applications, the lowest wage rate at which agents can find jobs, and $s_{\max}$ is symmetrically the greatest possible s_i .⁴ Compared with the previous chapter, this amounts to assuming that the production function is fixed and linear, and it can be normalized as $f(\ell) = \ell$ for all $\ell \geq 0$, so we can forget about f in the definition of economies. An economy is now a list $E = (s_N, R_N)$, and the domain of all such economies is $\mathcal{E}$.

An allocation $z_N = (z_i)_{i \in N} \in X^N$ is feasible in economy $E = (s_N, R_N) \in \mathcal{E}$ if

$$\sum_{i \in N} c_i \leq \sum_{i \in N} s_i \ell_i.$$

Let $Z(E)$ denote the set of feasible allocations for E .

The following terminology will prove useful. For $s_i \in \mathbb{R}_+$ and $z_i = (\ell_i, c_i) \in X$, let $B(s_i, z_i) \subset X$ denote the *budget set* obtained with skill s_i and such that z_i is on the budget frontier:

$$B(s_i, z_i) = \{(\ell'_i, c'_i) \in X \mid c'_i - s_i \ell'_i \leq c_i - s_i \ell_i\}.$$

In the special case in which $s_i = 0$ and $c_i = 0$, we adopt the convention that

$$B(s_i, z_i) = \{(\ell'_i, c'_i) \in X \mid c'_i = 0, \ell'_i \geq \ell_i\}.$$

For $s_i \in \mathbb{R}_+$, $R_i \in \mathcal{R}$ and $z_i = (\ell_i, c_i) \in X$, let $B^*(s_i, R_i, z_i) \subset X$ denote the *implicit budget* at bundle z_i for any agent with characteristics (s_i, R_i) – that is, the budget set with slope s_i having the property that z_i is indifferent for R_i to the preferred bundle in that budget set:

$$B^*(s_i, R_i, z_i) = B(s_i, z'_i) \text{ for any } z'_i \text{ such that } z'_i I_i z_i \text{ and } z'_i \in \max_{|R_i} B(s_i, z'_i).$$

³ This broader class of compensation problems is the subject of Fleurbaey (2008).

⁴ Unvoluntary unemployment can be modeled by fixing $s_{\min} = 0$.

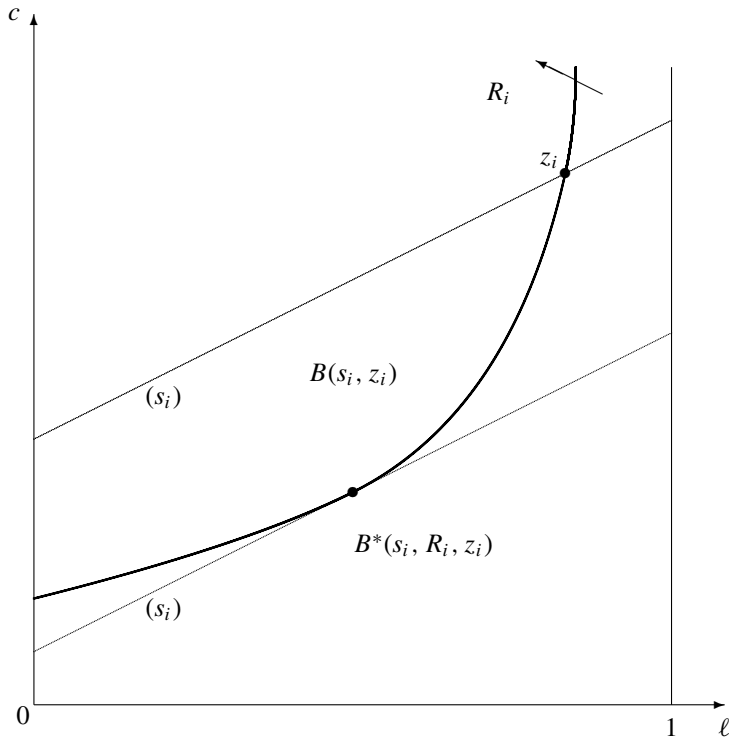


Figure 10.1. Budgets and implicit budgets

By strict monotonicity of preferences, this definition is unambiguous. Notice that bundle z_i need not belong to the implicit budget. Figure 10.1 illustrates these definitions. In the figures of this chapter, we adopt the convention that the slopes of budget lines are in parentheses below the line.

This new model does not require much rewriting of the axioms studied in the previous chapters. For instance, agents are now equals if they have the same preferences and skills. Also, independence of the feasible set amounts to requiring independence of the vector of skills.

10.3 COMPENSATION VERSUS RESPONSIBILITY: ETHICAL DILEMMAS

Let us now introduce the basic axioms capturing the two ethical objectives discussed in the previous section. Recall that, according to the objective of compensation, differential skills should not entail unequal individual outcomes. Consider two agents who have the same preferences but different skills. If one agent has a larger consumption, whereas they have the same labor time, one can argue that the preceding ethical goal is not fully satisfied and that, other things being equal, it would be a social improvement (or at least not a worsening)

to transfer an amount of the consumption good from the richer to the poorer agent. The resulting axiom is logically stronger than transfer among equals.

Axiom 10.1 Equal-Preferences Transfer

For all $E = (s_N, R_N) \in \mathcal{D}$, and $z_N = (\ell_N, c_N)$, $z'_N = (\ell'_N, c'_N) \in X^N$, if there exist $j, k \in N$ such that $R_j = R_k$, and $\Delta \in \mathbb{R}_{++}$ such that $\ell_j = \ell_k = \ell'_j = \ell'_k$,

$$c_j - \Delta = c'_j > c'_k = c_k + \Delta,$$

and for all $i \neq j, k$, $z_i = z'_i$, then $z'_N \mathbf{R}(E) z_N$.

Let us now turn to the ethical objective of responsibility. It implies that the various preferences displayed in the population should not be treated differently; that is, the agents should not receive any differential amount of resources on the pure basis that they have “good” or “bad” preferences. Therefore, agents with identical skills should ideally receive equal lump-sum transfers and then be left free to choose their preferred bundles in the same budget set with a slope equal to their skill. Therefore, the next axiom requires that a transfer of resources between two agents having the same skill and having received unequal lump-sum transfers should be a weak social improvement.

Before stating the axiom, we need to make the following formal point. Let us assume that agent j receives a lump-sum transfer and therefore maximizes his or her preferences over a budget with a slope equal to his or her skill, s_j . If $z_j = (\ell_j, c_j)$ is agent j 's chosen bundle, then $c_j - s_j \ell_j$, which may be negative, is equal to the lump-sum transfer this agent received and is a simple measure of the value of his or her budget. This measure allows us to introduce the notion of a “budget transfer” (corresponding to a transfer in lump-sum grants). If we move from an allocation where $z_j = (\ell_j, c_j)$ and $z_k = (\ell_k, c_k)$ to another allocation where $z'_j = (\ell'_j, c'_j)$ and $z'_k = (\ell'_k, c'_k)$, and if both agents maximize their welfare over budgets of slopes $s_j = s_k$, then we say that there has been a transfer of budget from j to k if $(c_j - s_j \ell_j) - (c'_j - s'_j \ell'_j) = (c'_k - s'_k \ell'_k) - (c_k - s_k \ell_k)$. Our main responsibility requirement can be phrased as follows: a budget transfer from an agent having a larger budget to another agent having the same skill should be viewed as a (weak) social improvement.

Axiom 10.2 Equal-Skill Transfer

For all $E = (s_N, R_N) \in \mathcal{D}$, $z_N = (\ell_N, c_N)$, $z'_N = (\ell'_N, c'_N) \in X^N$, $\Delta \in \mathbb{R}_{++}$, if there exist $j, k \in N$ such that $s_j = s_k$, $(c_j - s_j \ell_j) - \Delta = (c'_j - s'_j \ell'_j) > (c'_k - s'_k \ell'_k) = (c_k - s_k \ell_k) + \Delta$,

$$z_j \in \max_{|R_j} B(s_j, z_j), \quad z'_j \in \max_{|R_j} B(s_j, z'_j),$$

$$z_k \in \max_{|R_k} B(s_k, z_k), \quad z'_k \in \max_{|R_k} B(s_k, z'_k),$$

and for all $i \in N$, $i \neq j, k$, $z_i = z'_i$, then $z'_N \mathbf{R}(E) z_N$.

The first basic result of this chapter states that equal-preferences transfer and equal-skill transfer are incompatible with each other: it is impossible to guarantee at the same time that agents having identical preferences will be

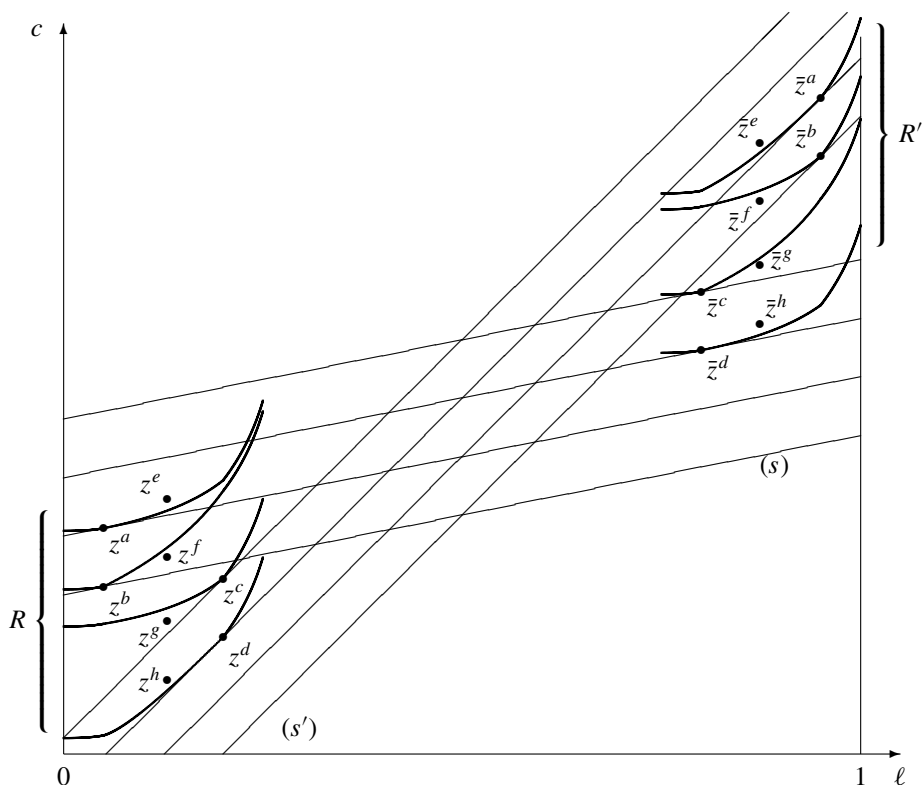


Figure 10.2. No SOF satisfies weak Pareto, equal-preferences transfer, and equal-skill transfer.

treated independently of their skills, whereas agents with identical skills will be treated, in terms of budgets, independently of their preferences.⁵

Theorem 10.1 *On the domain $\mathcal{E}$, no SOF satisfies weak Pareto, equal-preferences transfer, and equal-skill transfer.*

Proof. The proof is illustrated in Figure 10.2. Several budgets are depicted, associated to either skill s or s' , with $s < s'$, and several indifference curves are drawn, representing preferences R and R' .

Let us consider the four-agent economy $E = ((s, s, s', s'), (R, R', R, R'))$. Applying equal-skill transfer twice, first to the pair of agents (s, R) and (s, R') , and then to the pair (s', R) and (s', R') , we reach the conclusion that

$$(z^a, \bar{z}^d, z^d, \bar{z}^a) \mathbf{R}(E) (z^b, \bar{z}^c, z^c, \bar{z}^b).$$

⁵ This result, and much of this chapter, draws from Fleurbaey and Maniquet (2005). There has been a long literature on the compensation responsibility dilemma, surveyed in Fleurbaey and Maniquet (2010) and Fleurbaey (2008).

Applying equal-preferences transfer twice, first to the pair of agents (s, R) and (s', R) , and then to the pair (s, R') and (s', R') , we reach the conclusion that

$$(z^f, \bar{z}^g, z^g, \bar{z}^f) \mathbf{R}(E) (z^e, \bar{z}^h, z^h, \bar{z}^e).$$

Now, by weak Pareto, we get

$$(z^b, \bar{z}^c, z^c, \bar{z}^b) \mathbf{P}(E) (z^f, \bar{z}^g, z^g, \bar{z}^f)$$

and

$$(z^e, \bar{z}^h, z^h, \bar{z}^e) \mathbf{P}(E) (z^a, \bar{z}^d, z^d, \bar{z}^a)$$

which forms a cycle of strict social preference, a contradiction. ■

This simple illustration clearly demonstrates the tension between the two ethical objectives. Agent (s, R) , for instance, when consuming z_a , is considered relatively well-off when compared with agent (s', R) (who has the same preferences), according to the objective of compensation, but agent (s, R) is considered relatively disadvantaged when compared with agent (s, R') (who has the same skill), according to the objective of responsibility. This turns out to be the case for each of the four agents of that economy, with the consequence that the successive application of the axioms leads to a cycle.

The conflict between the two objectives is serious. We even find it impossible to combine them when we remove the egalitarian objective embedded in our two axioms, and when we replace them with anonymity axioms. Equal-preferences anonymity requires that permuting the bundles of two agents having the same preferences should not affect social welfare. Note that this axiom is not logically related to equal-preferences transfer.

Axiom 10.3 Equal-Preferences Anonymity

For all $E = (s_N, R_N) \in \mathcal{D}$, $z_N, z'_N \in X^N$, $j, k \in N$, if $R_j = R_k$, $z_j = z'_k$ and $z_k = z'_j$, and for all $i \in N$ such that $i \neq j, k$, $z_i = z'_i$, then $z'_N \mathbf{I}(E) z_N$.

Equal-skill anonymity requires that permuting the budgets (lump-sum grants) of two agents having the same skill, when they are precisely consuming their best bundles in their respective budgets, before and after the permutation, should not affect social welfare.

Axiom 10.4 Equal-Skill Anonymity

For all $E = (s_N, R_N) \in \mathcal{D}$, $z_N, z'_N \in X^N$, $j, k \in N$, if $s_j = s_k$, $B(s_j, z_j) = B(s_k, z'_k)$, $B(s_j, z'_j) = B(s_k, z_k)$, and

$$z_j \in \max_{|R_j} B(s_j, z_j), \quad z'_j \in \max_{|R_j} B(s_j, z'_j),$$

$$z_k \in \max_{|R_k} B(s_k, z_k), \quad z'_k \in \max_{|R_k} B(s_k, z'_k),$$

and for all $i \in N$, $i \neq j, k$, $z_i = z'_i$, then $z'_N \mathbf{I}(E) z_N$.

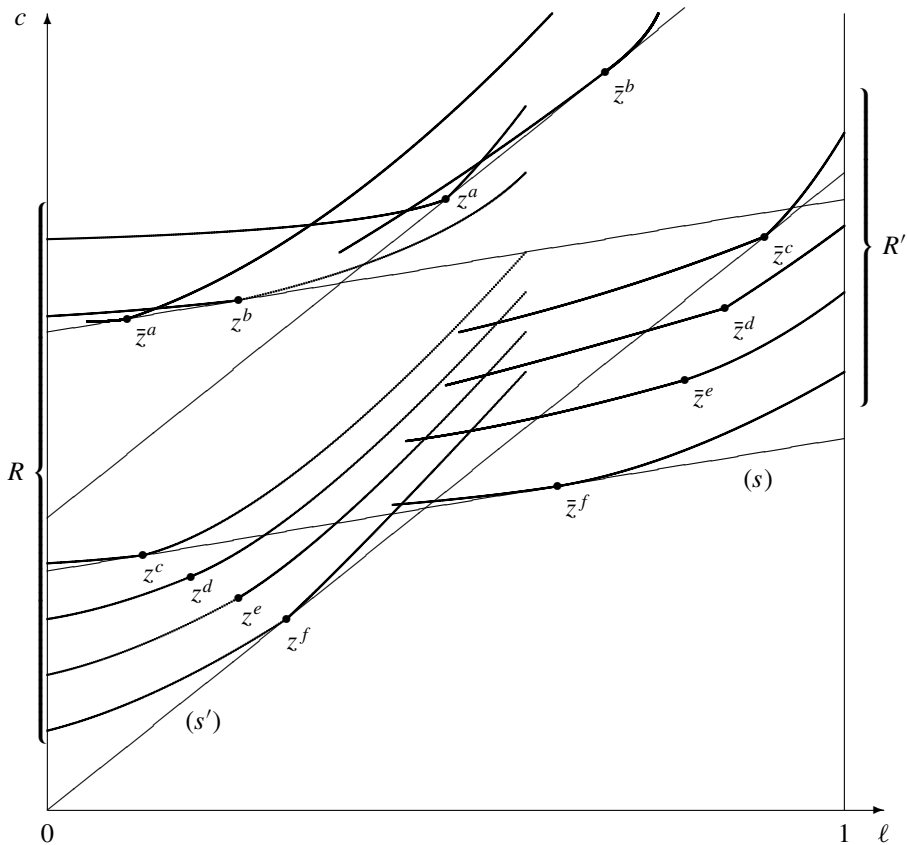


Figure 10.3. No SOF satisfies weak Pareto, equal-preferences anonymity, and equal-skill anonymity.

The second result of this chapter, unfortunately, confirms the difficulty, revealed by Theorem 10.1, of combining the objectives of compensation and responsibility.

Theorem 10.2 *On the domain $\mathcal{E}$, no SOF satisfies weak Pareto, equal-preferences anonymity, and equal-skill anonymity.*

Again, the proof can be graphical, involving an economy with two skill levels and two preferences $E = ((s, s, s', s'), (R, R', R, R'))$. We construct a cycle of allocations in Figure 10.3. Let us start with allocation $(z^b, \bar{z}^b, z^e, \bar{z}^e)$. By weak Pareto, $(z^a, \bar{z}^a, z^d, \bar{z}^d)$ is a strictly preferred allocation. By equal-preferences anonymity, it is indifferent to $(z^d, \bar{z}^e, z^a, \bar{z}^c)$, which, by equal-skill anonymity, is indifferent to $(z^d, \bar{z}^b, z^f, \bar{z}^e)$, itself indifferent to $(z^d, \bar{z}^b, z^f, \bar{z}^e)$, by equal-preferences anonymity. By weak Pareto, $(z^c, \bar{z}^a, z^e, \bar{z}^d)$ is strictly preferred, but

it turns out to be indifferent to the allocation we started with, $(z^b, \bar{z}^f, z^e, \bar{z}^d)$, by application of equal-skill anonymity.

10.4 WEAKENING THE BASIC AXIOMS

One possible way out of the preceding negative results consists of weakening the axioms. In this section, we study examples of such weakened axioms.

The first weakenings follow the idea that, in some economies, we know perfectly well what the optimal allocations should look like. Let us first apply this idea to the compensation objective. In an economy in which all agents have the same preferences, the only optimal allocations should be the Pareto efficient ones at which all agents are assigned bundles they deem equivalent. Indeed, no agent finds his or her bundle worse than that of any other agent, so that final outcomes can be claimed to have been equalized.

If we apply the same idea to the responsibility objective, we look at economies in which all agents have the same skill. In those economies, inequalities are justified, provided they come only from different choices. Consequently, letting agents maximize in the same budget set is optimal, and this is what laissez-faire prescribes. This leads us to the following two axioms.

Axiom 10.5 Equal-Welfare Selection

For all $E = (s_N, R_N) \in \mathcal{D}$, $z_N \in \max_{R(E)} Z(E)$, if for all $i, j \in N$, $R_i = R_j$, then for all $i, j \in N$, $z_i I_i z_j$.

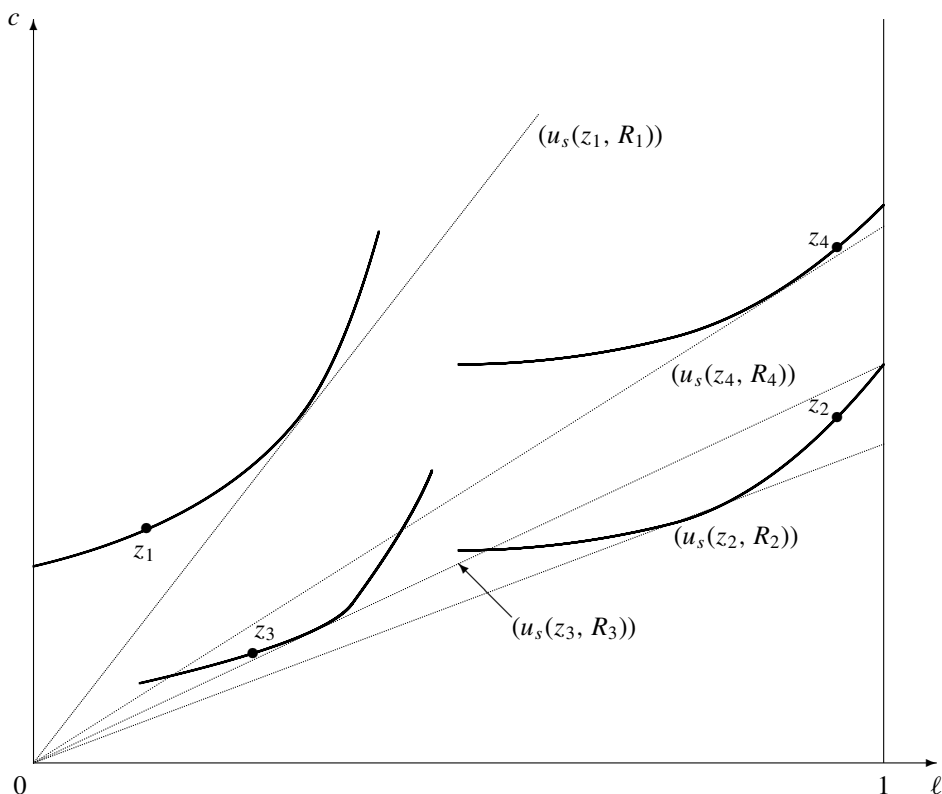
Axiom 10.6 Laissez-Faire Selection

For all $E = (s_N, R_N) \in \mathcal{D}$, $z_N \in \max_{R(E)} Z(E)$, if for all $i, j \in N$, $s_i = s_j$, then for all $i \in N$, $z_i \in \max_{R_i} B(s_i, (0, 0))$.

Equal-welfare selection is logically weaker than equal-preferences transfer,⁶ whereas laissez-faire selection is logically weaker than equal-skill transfer. Surprisingly, the duality between skills and preferences stops here. More precisely, we will show that, although it is easy to combine laissez-faire selection with equal-preferences transfer, the dual result does not hold (more precisely, combining weak pareto, equal-welfare selection, equal-skill transfer, and separability is impossible). We begin by exploring the former possibility. First, we define the skill-equivalent utility. It is the skill level that leaves the agent indifferent between his or her current bundle and being free to choose his or her labor time at that skill level. Formally, for $z_i \in X$, and $R_i \in R$,

$$u_s(z_i, R_i) = u \Leftrightarrow z_i I_i \max_{R_i} B(u, (0, 0)).$$

⁶ More precisely, it is logically implied by equal-preferences transfer under Pareto indifference and Hansson independence.

Figure 10.4. The s -equivalent leximin function

This well-being index is well defined only over bundles $z_i \in X$ such that $z_i R_i (0, 0)$. For $E = (s_N, R_N) \in \mathcal{E}$, let the set of admissible allocations $A(E)$ be defined as

$$A(E) = \{z_N \in Z(E) \mid z_i R_i (0, 0) \forall i \in N\}.$$

The following SOF is defined only over those allocations.

Social ordering function 10.1 s -Equivalent Leximin ($\mathbf{R}^{slex}$)

For all $E = (s_N, R_N) \in \mathcal{E}$, $z_N, z'_N \in A(E)$,

$$z_N \mathbf{R}^{slex}(E) z'_N \Leftrightarrow (u_s(z_i, R_i))_{i \in N} \geq_{lex} (u_s(z'_i, R_i))_{i \in N}.$$

This SOF is illustrated in Figure 10.4, representing a similar economy to the one used in the proof of the impossibility theorems 10.1 and 10.2. There are two skill levels, s and s' , and two preferences, R and R' ; the four agents have all the possible combinations of skill levels and preferences, so that $s_1 = s_2 = s < s_3 = s_4 = s'$, $R_1 = R_3 = R$ and $R_2 = R_4 = R'$. In allocation (z_1, z_2, z_3, z_4) , agent 1, the low-skilled agent with low willingness to work,

enjoys a larger satisfaction level than agent 3, who has the same preferences but a higher skill. On the other hand, among the agents with high willingness to work, the high-skilled one, agent 4, is assigned a better bundle than agent 2. According to the u_s well-being index, agent 2 is the worst-off agent, and agent 1 the best-off. At that allocation, R^{slex} would recommend redistribution from 1 and 4 to 2 and 3.

The preferred allocations for this SOF, among feasible allocations, are such that all the agents' skill-equivalent utilities are equal. This corresponds to the equal-wage equivalent allocation rule studied by Fleurbaey and Maniquet (1999).

Theorem 10.3 *On the domain $\mathcal{E}$, the s -equivalent leximin function $\mathbf{R}^{slex}$ satisfies strong Pareto, equal-preferences transfer, equal-preferences anonymity, laissez-faire selection, unchanged-contour independence, separation, replication, and independence of the feasible set. Conversely, if a SOF $\mathbf{R}$ satisfies strong Pareto, equal-preferences transfer, laissez-faire selection, unchanged-contour independence, and separation, then for all $E = (s_N, R_N) \in \mathcal{E}$, all $z_N, z'_N \in X^N$,*

$$\min_{i \in N} u_s(z_i, R_i) > \min_{i \in N} u_s(z'_i, R_i) \Rightarrow z_N \mathbf{P}(E) z'_N.$$

The proof makes use of the fact that strong Pareto, equal-preferences transfer, and unchanged-contour independence imply an absolute priority for the worse off (in the form of equal-preferences priority), as explained in Section 10.6. The other key element of the proof is to justify a preference for z_N over z'_N when, for some $j, k \in N$,

$$u_s(z'_j, R_j) > u_s(z_j, R_j) > u_s(z_k, R_k) > u_s(z'_k, R_k),$$

while for $i \neq j, k$, $z'_i = z_i$.

This is done in the following way. First, introduce two agents a and b (by separation) with the same skill s_0 between $u_s(z_k, R_k)$ and $u_s(z_j, R_j)$, and such that $R_a = R_j$, $R_b = R_k$. Let z_a, z_b be the bundles these two agents choose from the laissez-faire budget set $B(s_0, (0, 0))$. Observe that $z_j P_j z_a$ and $z_b P_k z_k$. By unchanged-contour independence and equal-preferences priority, agent a 's satisfaction can be raised a little while j is pulled down from z'_j to z_j , and agent b 's satisfaction can be lowered a little while k is moved from z'_k to a better bundle z''_k below z_k . By separation, one can focus on changes affecting a and b separately from the rest, and conclude that by laissez-faire selection, it would be at least as good to bring a and b back to their laissez-faire bundles. By separation again, one can remove a and b from the economy and conclude that moving j and k from z'_j, z'_k to z_j, z''_k is a weak improvement. By strong Pareto, moving them to z_j, z_k is a strict improvement. After this argument is made, it is easy to reach the conclusion of the theorem, in a similar fashion as has been done in the characterizations of Chapter 5.

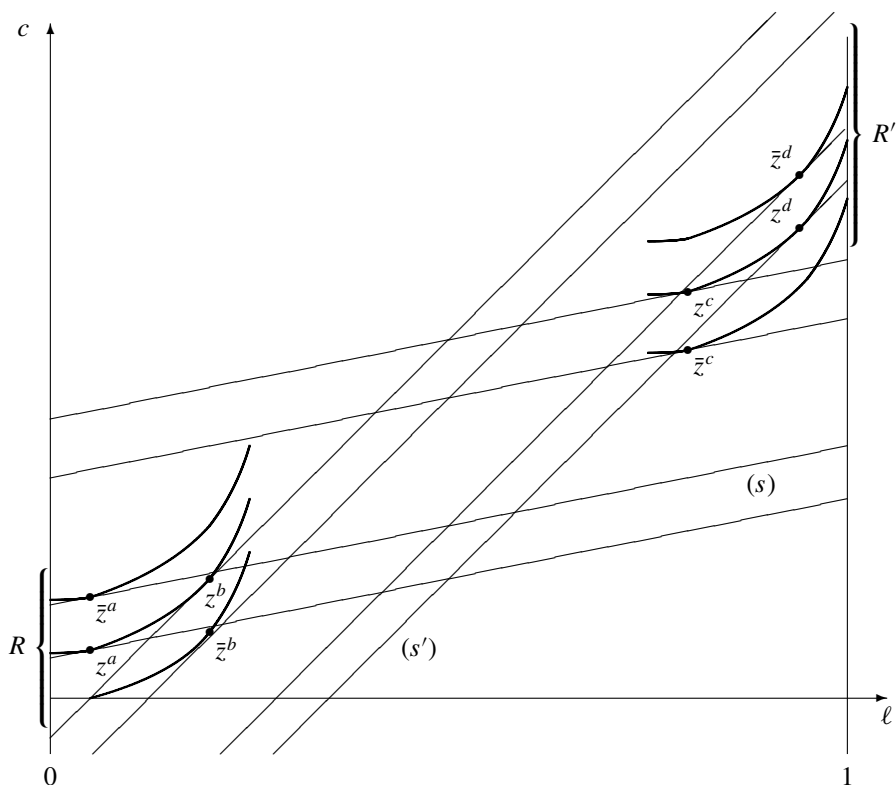


Figure 10.5. No SOF satisfies weak Pareto, equal-skill transfer, equal-welfare selection, and separability.

On the other hand, if we try to combine equal-welfare selection with equal-skill transfer, which is the dual to the combination of laissez-faire selection and equal-preferences transfer, we face the following difficulty.

Theorem 10.4 *On the domain $\mathcal{E}$, no SOF satisfies weak Pareto, equal-skill transfer, equal-welfare selection, and separability.*

Proof. The economy is $E = ((s, s', s, s'), (R, R, R', R'))$ but we will also consider the variants $E' = ((s, s', s, s'), (R, R, R, R))$ and $E'' = ((s, s', s, s'), (R', R', R', R'))$. The proof is illustrated in Figure 10.5.

By weak Pareto and equal-welfare selection, the unique best allocation in $Z(E')$ for $\mathbf{R}(E')$ is (z^a, z^b, z^a, z^b) . Therefore,

$$(z^a, z^b, z^a, z^b) \mathbf{P}(E') (\bar{z}^a, \bar{z}^b, z^a, z^b).$$

By separability, this is equivalent to

$$(z^a, z^b, z^c, z^d) \mathbf{P}(E) (\bar{z}^a, \bar{z}^b, z^c, z^d).$$

By similar reasoning, as the unique best allocation in $Z(E'')$ for $\mathbf{R}(E'')$ is (z^c, z^d, z^c, z^d) , one deduces that

$$(\bar{z}^a, \bar{z}^b, z^c, z^d) \mathbf{P}(E) (\bar{z}^a, \bar{z}^b, \bar{z}^c, \bar{z}^d).$$

By transitivity,

$$(z^a, z^b, z^c, z^d) \mathbf{P}(E) (\bar{z}^a, \bar{z}^b, \bar{z}^c, \bar{z}^d).$$

By equal-skill transfer,

$$(\bar{z}^a, z^b, \bar{z}^c, z^d) \mathbf{R}(E) (z^a, z^b, z^c, z^d)$$

and

$$(\bar{z}^a, \bar{z}^b, \bar{z}^c, \bar{z}^d) \mathbf{R}(E) (\bar{z}^a, z^b, \bar{z}^c, z^d),$$

implying by transitivity:

$$(\bar{z}^a, \bar{z}^b, \bar{z}^c, \bar{z}^d) \mathbf{R}(E) (z^a, z^b, z^c, z^d).$$

One therefore has a contradiction. ■

A similar impossibility would be reached if equal-skill transfer were replaced with equal-skill anonymity. The impossibility involves separability and vanishes in absence of this axiom. The axioms of strong Pareto, equal-skill transfer, and equal-welfare selection are satisfied by any SOF that applies the leximin criterion to indices of well-being defined as

$$\max \{u_{R_N}(z) \mid z \in B^*(s_i, R_i, z_i)\},$$

where u_{R_N} is an increasing continuous utility function that depends on the profile R_N and satisfies the property that when there is R_0 such that for all $i \in N$, $R_i = R_0$, then u_{R_N} is a representation of R_0 (for instance, u_{R_N} can be the utility representation of an "average" of agents' preferences).

We now study another possible weakening of equal-skill transfer, proposed by Valletta (2009a) in a slightly different context. Let us define the following partial order on the set $\mathcal{R}$ of preferences. For $R_i, R'_i \in \mathcal{R}$, we say that R_i is at least as industrious as R'_i , denoted $R_i \succsim^{\text{MI}} R'_i$, if and only if for every possible budget set, the best bundle of R_i involves at least as much labor (and, therefore, at least as much consumption) as that of R'_i . Formally,

$$R_i \succsim^{\text{MI}} R'_i \Leftrightarrow \forall s \in S, x \in X, \max_{|R_i} B(s, x) \geq \max_{|R'_i} B(s, x).$$

The "at least as industrious as" relationship is, of course, not complete.

The next axiom requires applying equal-skill transfer to pairs of agents such that the relatively richer agent is also more industrious. That axiom is consistent with the idea that consumption inequality, between two agents having the same skill, should be bounded above by what the difference in labor time justifies; that is, if $s_j = s_k$, $R_j \succsim^{\text{MI}} R_k$, and $c_j - c_k > s_j(\ell_j - \ell_k)$, then j has a relatively too-large consumption level, and a budget transfer from j to k is desirable. What the axiom does not require, contrary to equal-skill transfer, is that the

optimal difference in consumption be exactly the one related to the difference in labor time: $c_j - c_k = s_j(\ell_j - \ell_k)$. Consequently, the axiom also has a small compensation flavor, and that is why it is compatible with equal-preferences transfer.

Axiom 10.7 $\succsim^{\text{MI}}$ -Equal-Skill Transfer

For all $E = (s_N, R_N) \in \mathcal{D}$, $z_N = (\ell_N, c_N)$, $z'_N = (\ell'_N, c'_N) \in X^N$, $\Delta \in \mathbb{R}_{++}$, if there exist $j, k \in N$ such that $s_j = s_k$, $R_j \succsim^{\text{MI}} R_k$, $(c_j - s_j \ell_j) - \Delta = (c'_j - s'_j \ell'_j) > (c'_k - s'_k \ell'_k) = (c_k - s_k \ell_k) + \Delta$,

$$z_j \in \max_{R_j} B(s_j, z_j), \quad z'_j \in \max_{R_j} B(s_j, z'_j),$$

$$z_k \in \max_{R_k} B(s_k, z_k), \quad z'_k \in \max_{R_k} B(s_k, z'_k),$$

and for all $i \in N$, $i \neq j, k$, $z_i = z'_i$, then $z'_N \mathbf{R}(E) z_N$.

The following SOF satisfies equal-preferences transfer and $\succsim^{\text{MI}}$ -equal-skill transfer. It works by applying the leximin to the following $s_{\min}$ -equivalent utility. It equals the lump-sum transfer that leaves the agent indifferent between his or her current bundle and being free to choose his or her labor time in a budget set defined by this lump-sum transfer and the minimal skill level. Formally, for $z_i \in X$, and $R_i \in \mathcal{R}$,

$$u_{s_{\min}}(z_i, R_i) = u \Leftrightarrow z_i I_i \max_{R_i} B(s_{\min}, (0, u)).$$

Social ordering function 10.2 $s_{\min}$ -Equivalent Leximin ($\mathbf{R}^{s_{\min} \text{lex}}$)

For all $E = (s_N, R_N) \in \mathcal{E}$, $z_N, z'_N \in X^N$,

$$z_N \mathbf{R}^{s_{\min} \text{lex}}(E) z'_N \Leftrightarrow (u_{s_{\min}}(z_i, R_i))_{i \in N} \geq_{\text{lex}} (u_{s_{\min}}(z'_i, R_i))_{i \in N}.$$

This SOF is illustrated in Figure 10.6, using the same economy and the same allocation as in Figure 10.4. As with $\mathbf{R}^{s_{\min} \text{lex}}$, agent 1 is considered better-off than agent 3, and agent 4 better-off than agent 2, but any SOF satisfying equal-preferences transfer would consider this. The interesting observation is that agent 1 is now considered worse-off than agent 4 and agent 3 worse-off than agent 2. A social-welfare-improving reform would now ask agent 4 to contribute more and agent 1 less than in the previous case. Compared with $\mathbf{R}^{s_{\min} \text{lex}}$, $\mathbf{R}^{s_{\min} \text{lex}}$ is less favorable to the hardworking agents.

We have the following axiomatization result, the proof of which is in the appendix.

Theorem 10.5 On the domain $\mathcal{E}$, the $s_{\min}$ -equivalent leximin function $\mathbf{R}^{s_{\min} \text{lex}}$ satisfies strong Pareto, equal-preferences transfer, equal-preferences anonymity, $\succsim^{\text{MI}}$ -equal-skill transfer, unchanged-contour independence, separation, replication and independence of the feasible set. Conversely, if a SOF $\mathbf{R}$ satisfies strong Pareto, equal-preferences transfer, $\succsim^{\text{MI}}$ -equal-skill transfer, unchanged-contour independence, and separation, then for all $E = (s_N, R_N) \in \mathcal{E}$, all $z_N, z'_N \in X^N$,

$$\min_{i \in N} u_{s_{\min}}(z_i, R_i) > \min_{i \in N} u_{s_{\min}}(z'_i, R_i) \Rightarrow z_N \mathbf{P}(E) z'_N.$$

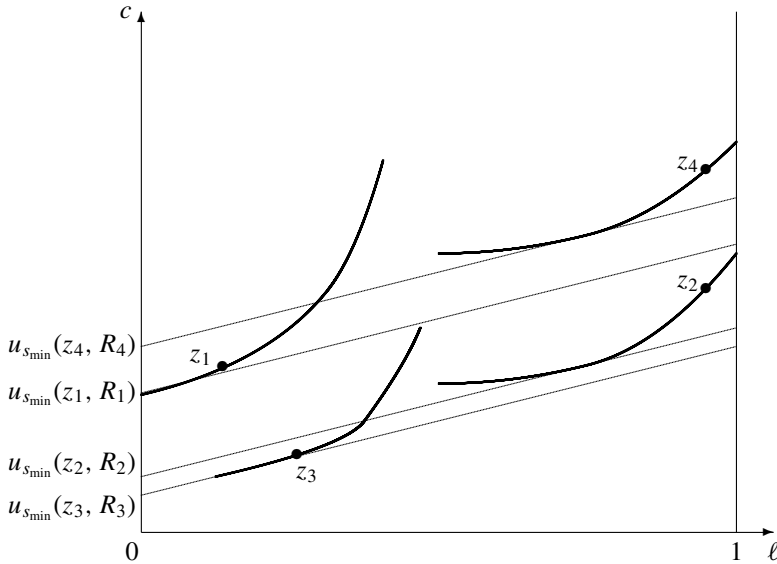


Figure 10.6. The $s_{\min}$ -equivalent leximin function

The proof is similar to that of Theorem 10.3 and again relies on the fact that equal-preferences priority follows from strong Pareto, equal-preferences transfer, and unchanged-contour independence (see Section 10.6). One also proves that there must be a preference for z_N over z'_N when for some $j, k \in N$,

$$u_{s_{\min}}(z'_j, R_j) > u_{s_{\min}}(z_j, R_j) > u_{s_{\min}}(z_k, R_k) > u_{s_{\min}}(z'_k, R_k),$$

whereas for $i \neq j, k$, $z'_i = z_i$.

This is done in the following way. First, introduce two agents a and b (by separation) with skill equal to $s_{\min}$, and with $s_{\min}$ -equivalent utilities between $u_{s_{\min}}(z_j, R_j)$ and $u_{s_{\min}}(z'_k, R_k)$. Their preferences are constructed so that the indifference curve of a (respectively, b) does not cross $I(z_j, R_j)$ (respectively, $I(z_k, R_k)$). Therefore, by unchanged-contour independence and equal-preferences priority, agent a 's satisfaction can be raised a little while j is pulled down from z'_j to z_j , and agent b 's satisfaction can be lowered a little while k is moved from z'_k to a better bundle z''_k below z_k . Moreover, their preferences must be such that for any budget with slope $s_{\min}$ or more, a chooses to work full time (this is obtained if a has linear preferences with a slope of indifference curves slightly less than $s_{\min}$). Therefore his preferences dominate those of b for $\succsim^{\text{MI}}$, making it possible to invoke $\succsim^{\text{MI}}$ -equal-skill transfer and bring a and b back to their initial bundles. By separation, they can then be removed from the economy. In conclusion, one obtains that moving j and k from z'_j, z'_k to z_j, z''_k is acceptable. By strong Pareto, moving them to z_j, z_k is then a strict improvement.

10.5 COMPROMISE AXIOMS

Another way of avoiding the negative result of Section 10.3 is by defining axioms that combine the objectives of compensation for low skill and responsibility for preferences. We give an example of such an axiom in this section.⁷ This axiom is consistent with Kolm's idea that members of a society should decide on a fraction $\tilde{\ell}$ of individual labor time such that incomes should be equalized among individuals working $\tilde{\ell}$ (see, e.g., Kolm 1996, 2004). More precisely, it requires that if two agents freely choose a labor time of $\tilde{\ell}$ but do not enjoy the same consumption level, then a transfer of consumption between them is a strict social improvement, provided they still end up freely choosing a labor time of $\tilde{\ell}$ after the transfer.

The compensation content of this axiom comes from the fact that the two agents may have different skills, so the larger consumption may originate in one agent having a higher skill. The axiom prevents the high-skilled agent from reaching a larger consumption level than the low-skilled one. The responsibility content of the axiom comes from the fact that it restricts its attention to agents having some particular preferences. Among such agents – that is, among agents choosing a labor time of $\tilde{\ell}$ – the objective of the axiom is to equalize consumption, so if the two agents have the same skill, a budget transfer between them is a social improvement.

Axiom 10.8 $\tilde{\ell}$ -Labor Transfer

For all $E = (s_N, R_N) \in \mathcal{D}$, and $z_N = (\ell_N, c_N)$, $z'_N = (\ell'_N, c'_N) \in X^N$, if there exist $j, k \in N$ and $\Delta \in \mathbb{R}_{++}$ such that $\ell_j = \ell_k = \ell'_j = \ell'_k = \tilde{\ell}$,

$$c_j - \Delta = c'_j > c'_k = c_k + \Delta,$$

and

$$z_j \in \max_{|R_j} B(s_j, z_j), \quad z'_j \in \max_{|R_j} B(s_j, z'_j),$$

$$z_k \in \max_{|R_k} B(s_k, z_k), \quad z'_k \in \max_{|R_k} B(s_k, z'_k),$$

and for all $i \neq j, k$, $z_i = z'_i$, then $z'_N \mathbf{R}(E) z_N$.

We do not have a theory about how to determine the value of $\tilde{\ell}$ (see Kolm 2004, part IV, for a detailed discussion of how societies may choose $\tilde{\ell}$). We can simply state that the larger the $\tilde{\ell}$, the more redistributive social preferences will be.

The key observation of this section is that there are SOFs that satisfy $\tilde{\ell}$ -labor transfer and equal-preferences transfer or $\tilde{\ell}$ -labor transfer and equal-skill transfer. The SOFs that satisfy $\tilde{\ell}$ -labor transfer and equal-preferences transfer are of the egalitarian-equivalent kind, and are somehow similar to the SOFs already defined in this chapter. A prominent example is the leximin SOF that measures individual well-being by the intersection of an agent's indifference curve with the vertical line of equation $\ell = \tilde{\ell}$.

⁷ This section is drawn from Fleurbaey and Maniquet (2011).

We concentrate in this section on the SOFs that satisfy $\tilde{\ell}$ -labor transfer and equal-skill transfer. They are of the Walrasian kind, but similar to egalitarian-equivalent SOFs, they also work by applying the leximin criterion to some index of well-being. Let us define this index. The novelty is that it depends on the skill of the agent. The level of well-being associated with a bundle is defined with respect to the budget that leaves the agent indifferent between this bundle and freely choosing in that budget the slope of which is now the real skill of the agent. When this budget has been identified, the value of the well-being index is the level of consumption that corresponds to a labor time of $\tilde{\ell}$ for that budget.

Formally, for $z_i \in X$, and $R_i \in R$,

$$u_{\tilde{\ell}}(z_i, s_i, R_i) = u \Leftrightarrow z_i \text{ } I_i \text{ } \max_{R_i} B(s_i, (\tilde{\ell}, u)).$$

This well-being index is nonnegative only over bundles $z_i \in X$ such that $z_i \text{ } R_i \text{ } \max_{R_i} B(s_i, (\tilde{\ell}, 0))$. For $E = (s_N, R_N) \in \mathcal{E}$, let the set of admissible allocations $A^{\tilde{\ell}}(E)$ be defined as

$$A^{\tilde{\ell}}(E) = \{z_N \in Z(E) \mid \forall i \in N, z_i \text{ } R_i \text{ } \max_{R_i} B(s_i, (\tilde{\ell}, 0))\}.$$

The following SOF is defined only over those allocations.⁸

Social ordering function 10.3 $\tilde{\ell}$ -Egalitarian Walrasian Leximin ($\mathbf{R}^{\tilde{\ell}EW}$)

For all $E = (s_N, R_N) \in \mathcal{E}$, $z_N, z'_N \in A^{\tilde{\ell}}(E)$,

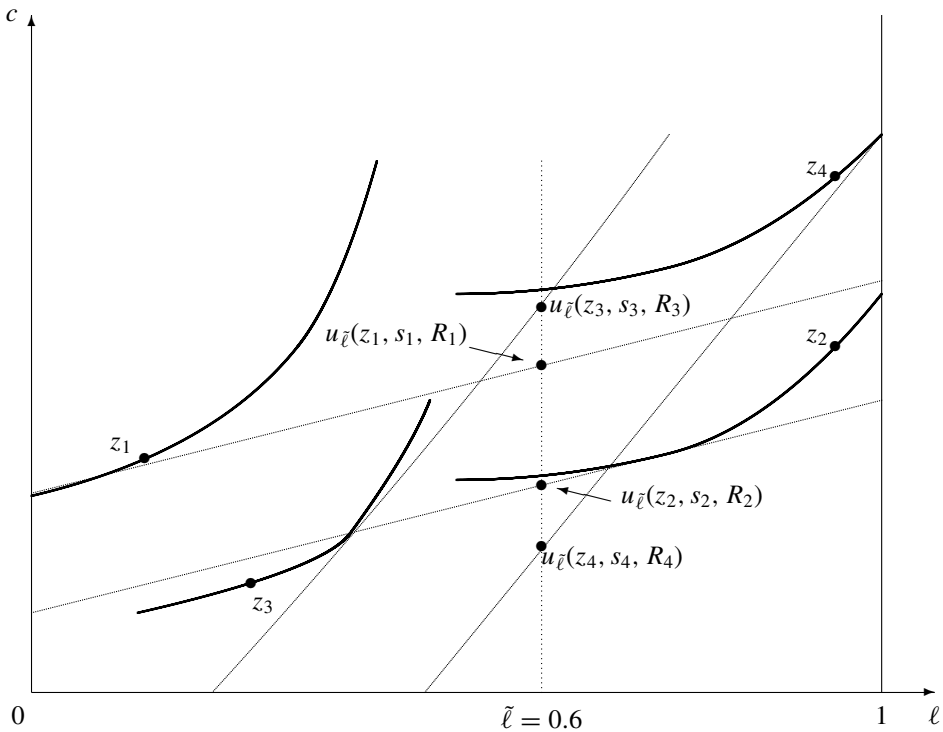
$$z_N \mathbf{R}^{\tilde{\ell}EW}(E) z'_N \Leftrightarrow (u_{\tilde{\ell}}(z_i, s_i, R_i))_{i \in N} \geq_{lex} (u_{\tilde{\ell}}(z'_i, s_i, R_i))_{i \in N}.$$

This SOF is illustrated in Figure 10.7, using the same economy and the same allocation as before. The well-being index is now computed by drawing implicit budgets having the same slope as the real skill of the agents. The different values of the index are read on the vertical axis corresponding to a labor time of $\tilde{\ell}$, which is chosen in the graph to be equal to 0.6.

We can see that the social evaluation of the agents' positions is drastically different from the previous ones. Agent 4 is now considered the worst off: the income agent 4 would earn by choosing a labor time of $\tilde{\ell}$ in his or her implicit budget is too low. A social welfare improving reform would benefit this agent. The other striking feature of the figure is that agent 1 is now considered worse off than agent 3. This shows that $\mathbf{R}^{\tilde{\ell}EW}$ benefits the hardworking high-skilled agents and the lazy low-skilled ones. This conclusion depends on the value of $\tilde{\ell}$ and is valid only for intermediate values. For $\tilde{\ell}$ sufficiently low, the worse-off agents would become 3 and 4, the two high-skilled agents, whereas for large values of $\tilde{\ell}$, the worse-off would become 1 and 2, the two low-skilled agents.⁹

⁸ Allowing for negative values of the index, it is easy to extend this SOF over all X^N . We do not do it here because the characterization of this SOF that is proposed in Section 10.6 works only for the admissible allocations.

⁹ The allocation rules that select the best allocations for $\mathbf{R}^{\tilde{\ell}EW}$, for given values of $\tilde{\ell}$, were introduced by Kolm (1996) and axiomatized by Fleurbaey and Maniquet (1996a) and Maniquet (1998). The

Figure 10.7. The $\tilde{\ell}$ -egalitarian Walrasian leximin

The $\mathbf{R}^{\tilde{\ell}\text{EW}}$ SOF satisfies strong Pareto, equal-skill transfer, equal-skill anonymity, $\tilde{\ell}$ -labor transfer, unchanged-contour independence, separation, and replication. On the other hand, it does not satisfy independence of the feasible set, as the well-being index that it uses depends on the skills of the agents, so that changing the skill profile affects the social ranking. Importantly, many other SOFs satisfy the same list of axioms, and they do not all give absolute priority to the worst-off. We develop this issue in the next section.

10.6 FROM TRANSFER TO PRIORITY AXIOMS

All the SOFs that we have studied in this chapter up to now are leximin SOFs in a particular index of well-being. This is not a surprise, of course, given all the previous results, and given, especially, the discussion of Chapter 3. In this section, nonetheless, we look carefully at the results about infinite inequality aversion, as they are not all derived from the same combinations of axioms.

extreme cases in which $\tilde{\ell}$ are either 0 or 1 were discussed by Varian (1974) and Pazner and Schmeidler (1978a), respectively.

Combining equal-preferences transfer with Pareto axioms and either unchanged-contour independence or separation unsurprisingly leads to an axiom of equal-preferences priority. In contrast, a similar statement does not hold with equal-skill transfer. There are SOFs, indeed, that do satisfy strong Pareto, equal-skill transfer, and strong robustness requirements, and are not of the leximin type. Here is an example. Let $\alpha \leq 1$, $\alpha \neq 0$. The SOF $\mathbf{R}$ defined by:

for all $E \in \mathcal{E}$, $z_N, z'_N \in A^{\bar{\ell}}(E)$,

$$z_N \mathbf{R}(E) z'_N \Leftrightarrow \frac{1}{\alpha} \sum_{i \in N} (u_{\bar{\ell}}(z_i, R_i))^\alpha \geq \frac{1}{\alpha} \sum_{i \in N} (u_{\bar{\ell}}(z'_i, R_i))^\alpha,$$

satisfies strong Pareto, equal-skill transfer, equal-skill anonymity, unchanged-contour independence, separability, and replication.

Would it be promising to study non-leximin SOFs such as this one? This is doubtful. Indeed, for all values of α , the SOF just defined violates the simple transfer among equals axiom (recall that agents are equals in this model if they have both the same preferences and the same skill), and so would do any SOF satisfying Pareto indifference, equal-skill transfer, and unchanged-contour independence, as long as they do not satisfy equal-skill priority. This is what we prove now.

Axiom 10.9 Transfer among Equals

For all $E = (s_N, R_N) \in \mathcal{D}$, and $z_N = (\ell_N, c_N)$, $z'_N = (\ell'_N, c'_N) \in X^{|N|}$, $j, k \in N$, $\Delta \in \mathbb{R}_{++}$, if $s_j = s_k$ and $R_j = R_k$, $\ell_j = \ell_k = \ell'_j = \ell'_k$, and

$$c_j - \Delta = c'_j > c'_k = c_k + \Delta,$$

and for all $i \neq j, k$, $z_i = z'_i$, then $z'_N \mathbf{R}(E) z_N$.

Axiom 10.10 Equal-Skill Priority

For all $E = (s_N, R_N) \in \mathcal{D}$, $z_N = (\ell_N, c_N)$, $z'_N = (\ell'_N, c'_N) \in X^N$, if there exist $j, k \in N$ such that $s_j = s_k$, $(c_j - s_j \ell_j) > (c'_j - s'_j \ell'_j) > (c'_k - s'_k \ell'_k) > (c_k - s_k \ell_k)$,

$$z_j \in \max_{|R_j} B(s_j, z_j), \quad z'_j \in \max_{|R_j} B(s_j, z'_j),$$

$$z_k \in \max_{|R_k} B(s_k, z_k), \quad z'_k \in \max_{|R_k} B(s_k, z'_k),$$

and for all $i \neq j, k$, $z_i = z'_i$, then $z'_N \mathbf{R}(E) z_N$.

Lemma 10.1 On the domain $\mathcal{E}$, if a SOF $\mathbf{R}$ satisfies Pareto indifference, transfer among equals, equal-skill transfer, unchanged-contour independence, and separation, then it satisfies equal-skill priority.¹⁰

¹⁰ Separation can be weakened into separability in this lemma if there are at least four agents in the economy. Also, unchanged-contour independence can be dropped altogether if transfer among equals is strengthened into nested-contour transfer.

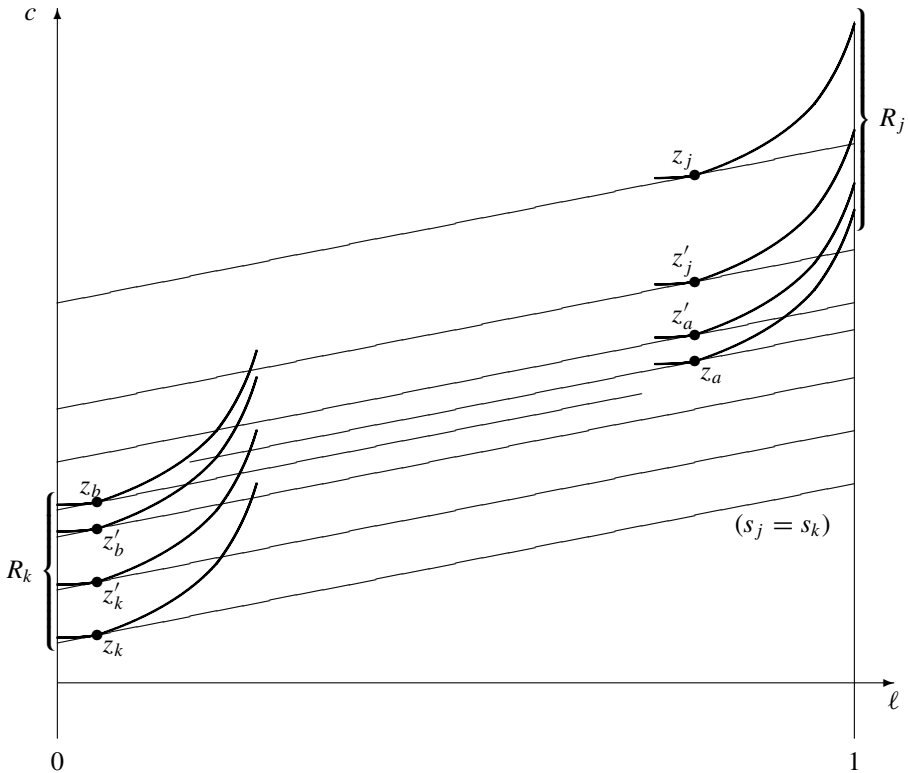


Figure 10.8. Pareto indifference, transfer among equals, equal-skill transfer, unchanged-contour independence, and separation imply equal-skill priority.

Proof. The proof, which bears some similarity with the proof of Theorem 10.5, is illustrated in Figure 10.8 which features four agents: j, k such that $s_j = s_k$, and a, b such that $(s_a, R_a) = (s_j, R_j)$ and $(s_b, R_b) = (s_k, R_k)$.

First, with the same reasoning as for Theorem 3.1, one sees that Pareto indifference, transfer among equals, and unchanged-contour independence imply priority among equals.

Axiom 10.11 Priority among Equals

For all $E = (s_N, R_N) \in \mathcal{D}$, and $z_N = (\ell_N, c_N)$, $z'_N = (\ell'_N, c'_N) \in X^{|N|}$, $j, k \in N$, $\Delta \in \mathbb{R}_{++}$, if $s_j = s_k$ and $R_j = R_k$, $\ell_j = \ell_k = \ell'_j = \ell'_k$, and

$$c_j > c'_j > c'_k > c_k,$$

and for all $i \neq j, k$, $z_i = z'_i$, then $z'_N \mathbf{R}(E) z_N$.

Then, one can apply priority among equals between j and a and between k and b (see Figure 10.8) to derive that

$$(z'_j, z'_k, z'_a, z'_b) \mathbf{R}((s_j, s_k, s_j, s_k), (R_j, R_k, R_j, R_k)) (z_j, z_k, z_a, z_b).$$

By equal-skill transfer applied to a and b ,

$$(z'_j, z'_k, z_a, z_b) \mathbf{R}((s_j, s_k, s_j, s_k), (R_j, R_k, R_j, R_k)) (z'_j, z'_k, z'_a, z'_b).$$

By transitivity,

$$(z'_j, z'_k, z_a, z_b) \mathbf{R}((s_j, s_k, s_j, s_k), (R_j, R_k, R_j, R_k)) (z_j, z_k, z_a, z_b).$$

By separation,

$$(z'_j, z'_k) \mathbf{R}((s_j, s_k), (R_j, R_k)) (z_j, z_k).$$

By separation again, reintroducing the other agents $i \neq j, k$ for whom $z'_i = z_i$, one obtains $z'_N \mathbf{R}(E) z_N$. ■

Equal-skill transfer seemed to be compatible with a noninfinite rate of inequality aversion, but the addition of the basic requirement of transfer among equals forces us to reintroduce an infinite inequality aversion. Lemma 10.1 confirms that only leximin types of SOFs are left when one tries to combine axioms of efficiency, fairness, and robustness. Indeed, we get the following result.

Theorem 10.6 *On the domain $\mathcal{E}$, if a SOF $\mathbf{R}$ satisfies strong Pareto, transfer among equals, equal-skill transfer, $\tilde{\ell}$ -labor transfer, unchanged-contour independence, and separation, then for all $E = (s_N, R_N) \in \mathcal{E}$, all $z_N, z'_N \in X^N$,*

$$\min_{i \in N} u_{\tilde{\ell}}(z_i, s_i, R_i) > \min_{i \in N} u_{\tilde{\ell}}(z'_i, s_i, R_i) \Rightarrow z_N \mathbf{P}(E) z'_N.$$

We simply illustrate the key part of the easy proof in Figure 10.9. It consists of showing that if for two agents $j, k \in N$,

$$u_{\tilde{\ell}}(z'_j, s_j, R_j) > u_{\tilde{\ell}}(z_j, s_j, R_j) > u_{\tilde{\ell}}(z_k, s_k, R_k) > u_{\tilde{\ell}}(z'_k, s_k, R_k),$$

whereas for $i \neq j, k$, $z'_i = z_i$, the allocation z_N is better than z'_N .

Introduce agents a, b such that $s_a = s_j$ and $s_b = s_k$. By Lemma 10.1, we know that we can apply equal-skill priority. Doing so between j and a , as well as between k and b , one sees that

$$(z_j, z'_k, z_a, z_b) \mathbf{R}((s_j, s_k, s_j, s_k), (R_j, R_k, R_a, R_b)) (z'_j, z'_k, z'_a, z'_b).$$

By $\tilde{\ell}$ -labor transfer applied to a and b ,

$$(z_j, z'_k, z'_a, z'_b) \mathbf{R}((s_j, s_k, s_j, s_k), (R_j, R_k, R_a, R_b)) (z_j, z'_k, z''_a, z''_b).$$

By transitivity,

$$(z_j, z'_k, z'_a, z'_b) \mathbf{R}((s_j, s_k, s_j, s_k), (R_j, R_k, R_a, R_b)) (z'_j, z'_k, z'_a, z'_b).$$

By separation,

$$(z_j, z'_k) \mathbf{R}((s_j, s_k), (R_j, R_k)) (z'_j, z'_k).$$

By strong Pareto,

$$(z_j, z_k) \mathbf{P}((s_j, s_k), (R_j, R_k)) (z'_j, z''_k),$$

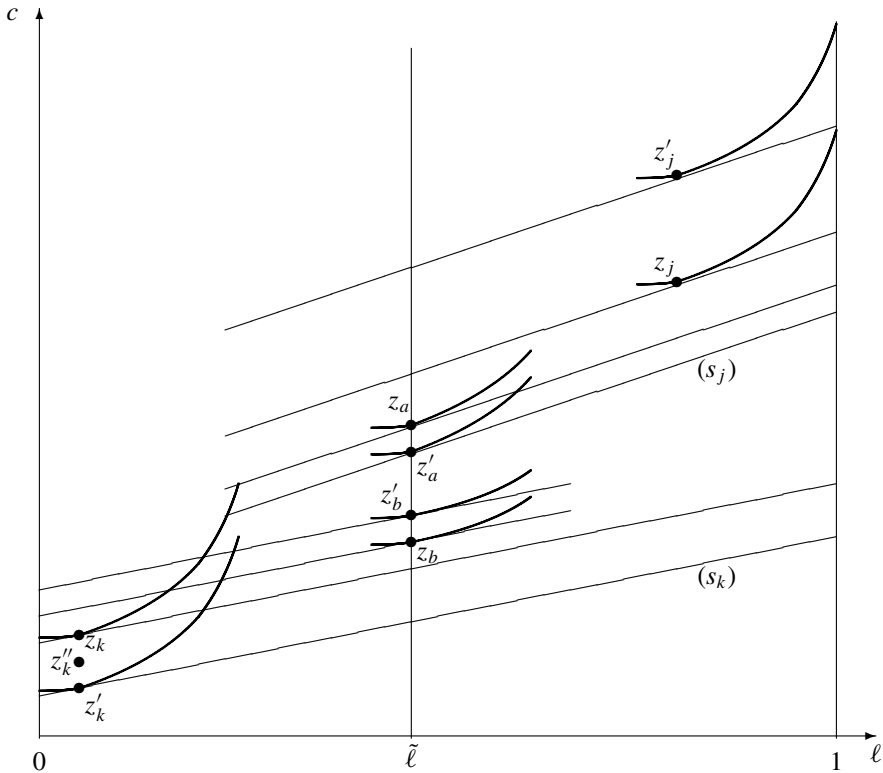


Figure 10.9. Illustration of the proof of Theorem 10.6

so

$$(z_j, z_k) \mathbf{P}((s_j, s_k), (R_j, R_k)) (z'_j, z'_k).$$

By separation again, reintroducing the other agents (with unchanged bundles in z_N and z'_N), one obtains $z_N \mathbf{P}(E) z'_N$.

10.7 CONCLUSION

Several families of SOFs end up being characterized in this chapter. Table 10.1 summarizes the relative merits of each of them.

Again, this chapter has illustrated what can be expected from the SOF approach: identify the possible trade-offs between fairness axioms, capturing ethical views on the resource allocation problem at hand; and deduce the SOF that satisfies some relevant set of fairness properties, together with efficiency and robustness axioms. The main ethical conflict in the model we have studied here is the one between compensation for low skills and responsibility for individual preferences. We saw that those conflicting axioms end up justifying

Table 10.1. *Properties of the SOFs*

	$\mathbf{R}^{slex}$	$\mathbf{R}^{s_{min}lex}$	$\mathbf{R}^{\tilde{\ell}EW}$
Strong Pareto	+	+	+
Transfer among equals	+	+	+
Equal-preferences transfer	+	+	−
Equal-preferences selection	+	+	−
Equal-skill transfer	−	−	+
Laissez-faire selection	+	−	+
$\gtrsim^{MI}$ -Equal-skill transfer	−	+	+
$\tilde{\ell}$ -Labor transfer	−	−	+
Unchanged-contour independence	+	+	+
Separation	+	+	+
Replication	+	+	+
Independence of the feasible set	+	+	−

quite different SOFs. The $\mathbf{R}^{slex}$ SOF favors hardworking agents, the $\mathbf{R}^{s_{min}lex}$ SOF favors lazy agents, and they both favor low-skilled agents. The $\mathbf{R}^{\tilde{\ell}EW}$ SOF, for intermediate values of $\tilde{\ell}$, favors hardworking high-skilled or lazy low-skilled agents.

Given that the current model is the canonical model in which labor income taxation is discussed, we now investigate it further to identify which income tax scheme may be compatible with the maximization process of the SOFs defined here.